Jaco ter Hoeve

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Keywords: Collider phenomenology, proton structure, physics beyond the standard model, future colliders, neural simulation-based inference, open-source scientific software

Professional Experience

since Oct 2024 — **Postdoctoral Research Associate**, *Higgs Centre, University of Edinburgh*

- Model-independent searches for new physics through the SMEFT
- Determination of the inner structure of the proton using advanced machine learning techniques
- Assessment of the physics potential of future colliders

Education

Oct 2020 – Sep 2024 — **PhD in Theoretical Physics**, *VU Amsterdam and Nikhef*

- Thesis: "Fingerprinting New Physics with Effective Field Theories"
  Advisors: Juan Rojo (VU), Wouter Verkerke (UvA)

Sep 2018 – Jul 2020 — **M.Sc. Theoretical Physics**, *Utrecht University*

- GPA: 8.0/10.0
- Thesis: "Matching between EFT and UV Complete Models" (at Nikhef)

Sep 2015 – Jun 2018 — **B.Sc. Physics & Astronomy**, *Utrecht University*

- Graduated cum laude, GPA: 8.8/10.0
- Thesis: "Renormalization Group Connected to Neural Networks"

Sep 2017 – Dec 2017 — **Erasmus exchange programme**, *University of Edinburgh*

- Obtained an average grade of 80%, 1st Class Honours

Publications

- H. Zhu, **J. ter Hoeve**, M. Kuusela, M. Ubiali, and L. Wasserman, "Semiparametric Estimation of Physics Parameters and Parton Distribution Functions," arXiv:2611.04267 [hep-ph].
- R. D. Ball and **J. ter Hoeve**, "Determining the Threshold Excess in Top-Quark Pair Production through a Global PDF Analysis," arXiv:2610.21479 [hep-ph].
- L. Vernazza, R. van Bijleveld, **J. ter Hoeve**, E. Laenen, C. Marinissen, and G. Wang, "Jet functions for next-to-leading power factorization," *PoS* **LL2026** (2026) 048, arXiv:2608.14382 [hep-ph].
- E. Celada, E. Hammou, **J. ter Hoeve**, and M. O. A. Thomas, "Descoped and Upscoped FCC-ee Running Scenarios in the SMEFT," arXiv:2608.05874 [hep-ph].
- T. Armadillo, E. Celada, **J. ter Hoeve**, F. Maltoni, L. Mantani, J. Rojo, A. N. Rossia, S. Tentori, M. O. A. Thomas, and E. Vryonidou, "New Physics Reach through Precision at Future Colliders: a Multi-Pronged Approach," arXiv:2604.16596 [hep-ph].
- R. Barrué *et al.*, "Proton Structure from Neural Simulation-Based Inference at the LHC," arXiv:2604.13157 [hep-ph].
- R. D. Ball, **J. ter Hoeve**, and R. Stegeman, "A determination of the top mass from a global PDF analysis," *JHEP* **08** (2026) 159, arXiv:2603.28865 [hep-ph].

- R. D. Ball, T. Giani, F. Hekhorn, **J. ter Hoeve**, T. R. Rabemananjara, J. Rojo, R. Stegeman, and M. Ubiali, "Parton distribution functions and theory parameters: an NNPDF perspective," arXiv:2602.24005 [hep-ph].
- **J. ter Hoeve**, L. Mantani, J. Rojo, A. N. Rossia, and E. Vryonidou, "Higgs trilinear coupling in the standard model effective field theory at the high luminosity LHC and the FCC-ee," *Phys. Rev. D* **112** no. 1, (2025) 013008, arXiv:2504.05974 [hep-ph].
- **J. ter Hoeve**, L. Mantani, J. Rojo, A. N. Rossia, and E. Vryonidou, "Connecting scales: RGE effects in the SMEFT at the LHC and future colliders," *JHEP* **06** (2025) 125, arXiv:2502.20453 [hep-ph].
- **J. ter Hoeve**, *Fingerprinting New Physics with Effective Field Theories*. PhD thesis, Vrije Universiteit Amsterdam, 2025. arXiv:2501.08818 [hep-ph].
- E. Celada, T. Giani, **J. ter Hoeve**, L. Mantani, J. Rojo, A. N. Rossia, M. O. A. Thomas, and E. Vryonidou, "Mapping the SMEFT at high-energy colliders: from LEP and the (HL-)LHC to the FCC-ee," *JHEP* **09** (2024) 091, arXiv:2404.12809 [hep-ph].
- **J. ter Hoeve**, E. Laenen, C. Marinissen, L. Vernazza, and G. Wang, "Region analysis of QED massive fermion form factor," *JHEP* **02** (2024) 024, arXiv:2311.16215 [hep-ph].
- **J. ter Hoeve**, G. Magni, J. Rojo, A. N. Rossia, and E. Vryonidou, "The automation of SMEFT-assisted constraints on UV-complete models," *JHEP* **01** (2024) 179, arXiv:2309.04523 [hep-ph].
- H. La, A. Brokkelkamp, S. van der Lippe, **J. ter Hoeve**, J. Rojo, and S. Conesa-Boj, "Edge-induced excitations in Bi2Te3 from spatially-resolved electron energy-gain spectroscopy," *Ultramicroscopy* **254** (2023) 113841, arXiv:2305.03752 [cond-mat.mes-hall].
- R. Gomez Ambrosio, **J. ter Hoeve**, M. Madigan, J. Rojo, and V. Sanz, "Unbinned multivariate observables for global SMEFT analyses from machine learning," *JHEP* **03** (2023) 033, arXiv:2211.02058 [hep-ph].
- A. Brokkelkamp, **J. ter Hoeve**, I. Postmes, S. E. van Heijst, L. Maduro, A. V. Davydov, S. Krylyuk, J. Rojo, and S. Conesa-Boj, "Spatially Resolved Band Gap and Dielectric Function in Two-Dimensional Materials from Electron Energy Loss Spectroscopy," *J. Phys. Chem. A* **126** no. 7, (2022) 1255–1262, arXiv:2202.12572 [cond-mat.mtrl-sci].

Collaboration papers

- J. Altmann *et al.*, *ECFA Higgs, electroweak, and top Factory Study*, vol. 5/2025 of *CERN Yellow Reports: Monographs*. CERN, Geneva, 2025. arXiv:2506.15390 [hep-ex].
- **FCC** Collaboration, M. Benedikt *et al.*, "Future Circular Collider Feasibility Study Report: Volume 1, Physics, Experiments, Detectors," *Eur. Phys. J. C* **85** no. 12, (2025) 1468, arXiv:2505.00272 [hep-ex]. [Erratum: Eur.Phys.J.C 86, 844 (2026)].
- **FCC** Collaboration, M. Benedikt *et al.*, "Future Circular Collider Feasibility Study Report: Volume 2, Accelerators, Technical Infrastructure and Safety," *Eur. Phys. J. ST* **234** no. 19, (2025) 5713–6197, arXiv:2505.00274 [physics.acc-ph].
- **FCC** Collaboration, M. Benedikt *et al.*, "Future Circular Collider Feasibility Study Report," *Eur. Phys. J. ST* **234** no. 17, (2025) 5113–5383, arXiv:2505.00273 [physics.acc-ph]. [Erratum: Eur.Phys.J.ST 234, 33–44 (2025)].

Talks and seminars

Invited talks and seminars

Nov 2026	**Probing new physics through precision at future colliders**, *Theory Seminar, DESY Hamburg* (slides)
May 2026	**A Determination of the Top Mass from a Global PDF Analysis**, *LHC Top Working Group* (slides), *ATLAS PDF fit forum and CMS Top PAG meeting, CERN*
May 2025	**Connecting scales in the SMEFT at the LHC and future colliders**, *Seminar at DAMTP, University of Cambridge* (slides)
Dec 2024	**Mapping the SMEFT at future colliders, towards the ESPPU**, *Seminar at IPPP, Durham University*
May 2024	**Update on global SMEFT fits**, *Standard Model at the LHC, Rome* (slides)
Mar 2024	**Global SMEFT fits from (HL)-LHC to future colliders**, *Seminar at Radboud University, Nijmegen*
Feb 2024	**Likelihood-free inference for global SMEFT fits**, *1st COMETA General Meeting, Izmir* (slides)
Oct 2023	**Machine Learning opportunities for EFT analyses**, *LHC EFT Working Group, CERN (online)* (slides), (recording)
Oct 2023	**Likelihood learning theory in practice**, *LHC Precision Program, Benasque* (slides)
Aug 2023	**Interpreting HEP data in SMEFiT**, *Reinterpretation Forum, Durham* (slides)
Dec 2022	**Unbinned multivariate observables for global SMEFT analyses from machine learning**, *Reinterpretation Forum, CERN* (slides)
Sep 2022	**Towards an optimal global SMEFT fit with machine learning**, *Machine Learning at GGI, Florence* (slides)

Contributed talks

Sep 2026	**Proton structure from NSBI at the LHC**, *ML4Jets 2026, Vienna* (slides)
Jun 2026	**Global Fits at Future Colliders**, *HEFT 2026, Valencia* (slides)
Jun 2026	**Global Fits at Future Colliders**, *3rd COMETA General Meeting, Stockholm* (slides)
May 2026	**A Determination of the Top Mass from a Global PDF Analysis**, *DIS2026, Bologna* (slides)
Dec 2025	**Updates from the SMEFiT collaboration**, *LHC EFT Working Group (online)* (slides)
Jul 2025	**Towards a combined α_s and m_t determination from a global PDF analysis**, *EPS-HEP, Marseille* (slides)
Jul 2025	**Connecting scales: RGE effects in the SMEFT at the LHC and future colliders**, *EPS-HEP, Marseille* (slides)

Jan 2025	**Single particle extensions at FCC-ee**, *8th FCC Physics Workshop, CERN* (slides)
Jul 2024	**Mapping the SMEFT at High-Energy Colliders: from LEP and the (HL-)LHC to the FCC-ee**, *ICHEP, Prague* (slides)
Jun 2023	**Statistically optimal observables for global SMEFT fits**, *HEFT, Manchester* (slides)
May 2022	**Unbinned measurements in global SMEFT fits from machine learning**, *Learning to Discover, Paris-Saclay* (slides)
Jan 2022	**Charting Electron Energy Loss Spectra with Machine Learning**, *NWO Physics 2021, Veldhoven*
Nov 2021	**Optimally sensitive observables for global EFT fits**, *Dutch Physics Society (NNV) annual meeting, Lunteren* (slides)
Oct 2021	**Optimally sensitive observables for global EFT fits**, *Higgs 2021, Stony Brook University (online)* (slides)

Open-source software packages

- `SMEFiT`, global SMEFT analyses of particle physics data `https://smefit.science`
  Code coordinator and core developer: used for input to the European Strategy
- `NNPDF`, neural-network parton distribution functions `https://docs.nnpdf.science`
  Developer: used in for example JHEP 08 (2026) 159
- `ML4EFT`, simulation-based inference for unbinned EFT fits `https://lhcfitnikhef.github.io/ML4EFT`
  Lead developer: designed and wrote the package; used in JHEP 03 (2023) 033
- `EELSFitter`, ML tool for electron energy loss spectra `https://lhcfitnikhef.github.io/EELSfitter`
  Developer: used in J. Phys. Chem. A 126 (2022) 1255

Teaching and supervisory experience

Nov 2026	**Lecturer, Terascale Monte Carlo School 2026, DESY Hamburg** Lectured at the Terascale school organised by the Helmholtz Alliance (slides)
since Sep 2026	**Teaching Assistant, University of Edinburgh** Lead workshops in the course "Principles of Quantum Mechanics"
since Sep 2025	**PhD student co-supervision, University of Edinburgh** A. Pietraszkiewicz
Sep 2022 – Jun 2023	**Master thesis supervision, UvA and VU Amsterdam** P. Herbschleb (2023), "ML Optimal Observables in SMEFT"
2021 – 2023	**Bachelor thesis supervision, UvA and VU Amsterdam** W. Gauthier (2023), J. Bakker and D. Pelan (2021)
2021-2023	**Teaching Assistant, UvA and VU Amsterdam** Courses: Effective Field Theories (2023), Quantum Mechanics II (2021, 2022)
Nov 2016 – Jun 2020	**Teaching Assistant, Utrecht University** Courses: Black Holes (2019, 2020), Fluid Dynamics (2018, 2020), Data Analysis (2016, 2017)

Organisation of Conferences, Workshops and Seminars

ongoing	**Main organiser of the EFT Foundation and Tools workshop 2027** Obtained €15k funding from the Higgs Centre, Edinburgh, and the COMETA COST Action
Nov 2026	**Convener, International Workshop on Future Linear Colliders, Japan** Convener of the Global Analysis session
since May 2026	**Seminar organiser, Higgs Centre, University of Edinburgh** Organise and chair the Particle Physics Theory Seminars
since Feb 2026	**COMETA colloquia organiser** Organise the online COMETA colloquia series
Nov 2024	**Workshop co-organiser, Higgs Centre, University of Edinburgh** Organised a SMEFT workshop

Collaborations and Working Groups

- Member of the NNPDF collaboration
- Member of the SMEFiT collaboration
- Member of the LHC EFT Working Group
- Member of the LHC Top Working Group
- Member of the FCC-ee Working Group
- Member of the COMETA COST Action (COmprehensive Multiboson Experiment-Theory Action)

Refereeing

Referee for JHEP (3), Eur. Phys. J. C (1), Eur. Phys. J. A (1), SciPost (2)

Selected activities and outreach

Jun 2022 – Sep 2024	**NNV Subatomic Physics council member** Organised scientific events for the Dutch Subatomic Physics Society
Dec 2020 – Sep 2024	**Chair DRSTP PhD council** Organised scientific meetings and social/outreach events for theoretical physics PhD students in The Netherlands
Apr 2024	**Outreach talk, University of Amsterdam** Particle physics talk for Norwegian high school students
Mar 2022	**CERN Masterclass, Nikhef** Particle physics outreach workshop for high school students
May 2021	**Guest lecture, TU Delft** Topic: Electron Energy Loss Fitter with Machine Learning

Computer and natural languages

- Programming languages: Python, C++, Java, Fortran, Mathematica
- ML and scientific computing: PyTorch, TensorFlow, JAX
- Languages: Dutch (native), English (full professional proficiency)